

Module 2

Five-Task Comparison Matrix — complete all cells:

Prompt	Tool A name	Output summary	Acc. (1-5)	Rel. (1-5)	Fmt . (1-5)	Tool B name	Output summary	Acc. (1-5)	Rel. (1-5)	Fmt . (1-5)
1. What year was the Transformer architecture first introduced in a peer-reviewed paper, and what was that paper's exact title? Please provide an APA citation	Gemini	The Transformer architecture was first introduced in 2017 at the 31st Conference on Neural Information Processing Systems (NeurIPS 2017). The exact title of the paper is: "Attention Is All You Need" APA 7 Citation Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. Advances in Neural Information Processing Systems, 30, 5998–6008. Cited by: 255297	1	3	3	Claude	The Transformer architecture was introduced in the paper "Attention Is All You Need," published in 2017 at NeurIPS (then called NIPS — Neural Information Processing Systems). A clarification on "peer-reviewed": the paper first appeared on arXiv in June 2017, then was presented and published in the NeurIPS 2017 proceedings, which is peer-reviewed (NeurIPS uses a rigorous review process). So 2017 is the correct year for the peer-reviewed venue. APA citation (7th edition):	1	5	5

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							Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. <i>Advances in Neural Information Processing Systems</i>, 30, 5998–6008. A few notes if you're using this in formal academic work: Some style guides/citation managers list it as a conference paper rather than a journal-style entry, since NeurIPS proceedings function as both. If your citation manager pulls it as an arXiv preprint instead (arXiv:1706.03762), that's the earlier, non-peer-reviewed version — worth distinguishing if your context cares about that nuance.			

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							The author list order above is the canonical one from the original publication. Want me to also format this in a different style (MLA, Chicago, IEEE) or pull the arXiv version's citation for comparison?			
2.Summarize Ouyang abstract	Gemini	Just making AI models larger doesn't automatically mean they will listen to you or tell the truth. To fix this, researchers created InstructGPT by training OpenAI's GPT-3 using human feedback. First, humans showed the AI examples of good responses. Then, humans ranked different AI answers, and the model used reinforcement learning to improve based on those preferences. The results were massive: a smaller InstructGPT model with 100 times fewer parameters	1	2	3	Claude	Just making AI language models bigger doesn't automatically make them better at understanding what users actually want — bigger models can still give answers that are false, harmful, or unhelpful. This paper describes a fix: training models using human feedback. Researchers had people write examples of good responses, then rank different model outputs, and used both to fine-tune GPT-3 into a new version	1	3	3

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		actually outperformed the massive standard GPT-3. It was much better at following directions, more truthful, and less toxic, proving that human guidance is key to making AI helpful.					called InstructGPT. Remarkably, a much smaller InstructGPT model was preferred over the original, far larger GPT-3, while also being more truthful and less likely to produce harmful content — though it still makes occasional mistakes.			
2. Write a docstring	Gemini	Here is a Python function that calculates the mean, median, and standard deviation using the built-in statistics module, which handles the math efficiently and accurately. <pre>Python import statistics from typing import Dict, List def calculate_descriptive_stats(numbers: List[float]) -> Dict[str, float]: """Calculates the mean, median, and standard deviation of a list of numbers.</pre>	Unable to assess will come back to this	Unable to assess will come back to this	5		<pre>pythonimport statistics def compute_stats(numbers): """ Compute the mean, median, and standard deviation of a list of numbers. Args: numbers (list of int or float): A non-empty list of numeric values. Returns:</pre>	Unable to assess will come back to this	Unable to assess will come back to this	3

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		Args: numbers (List[float]): A list of numerical values (ints or floats). Returns: Dict[str, float]: A dictionary containing the keys 'mean', 'median', and 'standard_deviation'. Raises: ValueError: If the input list is empty or has fewer than 2 elements (required for sample standard deviation). """ if not numbers: raise ValueError("The input list cannot be empty.") if len(numbers) < 2: raise ValueError("Standard deviation requires at least two data points.")) return { "mean": statistics.mean(numbers),					dict: A dictionary with the following keys: - 'mean': The arithmetic mean of the numbers. - 'median': The median value of the numbers. - 'standard_deviation': The sample standard deviation of the numbers. Raises: ValueError: If the input list is empty. Example: <pre>>>> compute_stats([2, 4, 4, 4, 5, 5, 7, 9]) {'mean': 5.0, 'median': 4.5, 'standard_deviation': 2.138089935299395} """ if not numbers:</pre>			

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		<pre>"median": statistics.median(numbers), "standard_deviation": statistics.stdev(numbers), } # --- Usage Example --- if __name__ == "__main__": data = [10, 12, 23, 23, 16, 23, 21, 16] try: stats = calculate_descriptive_stats(data) print("Statistical Summary:") print("-" * 20) for key, value in stats.items(): print(f'{key.replace('_', ' ').title()}: {value:.4f}') except ValueError as e: print(f"Error: {e}")</pre> Output of the Example: Plaintext Statistical Summary: ----- Mean: 18.0000					<pre>raise ValueError("Input list must not be empty.") return { "mean": statistics.mean(numbers), "median": statistics.median(numbers), "standard_deviation": statistics.stdev(numbers) if len(numbers) > 1 else 0.0, } if __name__ == "__main__": data = [2, 4, 4, 4, 5, 5, 7, 9] result = compute_stats(data) print(result) A couple of notes: statistics.stdev computes the sample standard</pre>			

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		Median: 18.5000 Standard Deviation: 5.2644 Notes on this Implementation: statistics.stdev: This computes the <i>sample</i> standard deviation (divided by \$n - 1\$). If your use case requires the <i>population</i> standard deviation (divided by \$n\$), you can change it to statistics.pstdev. Edge Cases: The function includes basic guard rails to ensure the list isn't empty and contains enough data points to compute a meaningful standard deviation.					deviation (dividing by $n-1$). If you want the population standard deviation instead, swap it for statistics.pstdev. For a single-element list, sample standard deviation is undefined (division by zero), so I return 0.0 in that edge case rather than raising an error — let me know if you'd prefer it to raise instead.			
3. Declination letter	Gemini	Subject: Invitation to Speak at [Event Name] - [Your Name] Dear [Organizer's Name], Thank you so much for inviting me to speak at the upcoming [Event Name] on [Date]. I am incredibly honored that you	Coherence 3		3	Claude	Subject: Re: Speaking Invitation Dear [Name], Thank you so much for thinking of me for [event/occasion name]. I'm genuinely honored by the invitation, and it	4		3

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		reached out, and the theme of the event sounds absolutely fantastic. Unfortunately, I have a prior professional commitment on that date that I cannot reschedule, so I must regretfully decline the invitation this time around. It is always tough to pass up the opportunity to connect with such a wonderful community, and I am genuinely sorry to miss it. I would love to stay in touch and keep the door open for future opportunities. Please do keep me in mind for subsequent events or panel discussions down the road. In the meantime, I wish you and the entire team the absolute best of luck with a successful event! Warm regards, [Your Name] [Your Title]					sounds like a wonderful event. Unfortunately, I have a scheduling conflict on [date] that I'm unable to move, so I won't be able to join as a speaker this time. I'm disappointed to miss it, especially given how meaningful the topic is to me. That said, I'd love to stay in touch and would welcome the chance to be considered for future events. Please don't hesitate to reach out again — I'll do my best to make it work next time. In the meantime, if it would be helpful, I'm happy to suggest a few other speakers who might be a great fit for this year's event, or to support in some other way if			

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		[Your Contact Information/Website]					that's useful. Thank you again for the invitation, and I hope [event name] is a great success. Warm regards, [Your Name] <i>Drafted — fill in the bracketed placeholders ([Name], [event/occasion name], [date], [Your Name]) before sending. Let me know if you'd like it shorter, more formal, or tweaked in tone.</i>			
5 University disclosure pro and con	Gemini	Pro-Disclosure Implementing mandatory disclosure ensures academic integrity by providing total transparency into a student's actual skill level and authorship. Furthermore, it allows instructors to accurately assess critical thinking and adapt their teaching methods to the changing technological landscape. Con-Disclosure	3	4	2	Claude	In favor: Mandatory disclosure preserves the integrity of grading by letting instructors accurately assess a student's own competencies versus AI-assisted output. It also helps students develop honest habits around intellectual attribution that will matter in their	4	5	4

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		Enforcing a blanket disclosure policy is practically impossible due to the seamless integration of AI in everyday grammar checkers, search engines, and basic word processors. Additionally, punitive framing around disclosure may inadvertently discourage students from learning to use essential, industry-standard AI tools effectively and ethically.					future academic and professional careers. Against: Blanket disclosure requirements can be difficult to enforce or verify, since AI use exists on a spectrum from light editing assistance to wholesale generation, making consistent policy application impractical. Such rules may also penalize students who use AI as a legitimate learning aid (e.g., for brainstorming or grammar checks) in the same way as those who use it to bypass learning entirely, conflating very different behaviors under one policy.			

where

- Acc. = Accuracy (rated 1–5; factual correctness of claims you can independently verify)
- Rel. = Relevance (rated 1–5; how well the output addresses the prompt)
- Fmt. = Format (rated 1–5; appropriateness of structure, length, and presentation)